

# Methodology

## Project Workflow

### 1. Data Understanding

The transaction dataset was reviewed to understand its structure, available attributes, and potential indicators of fraudulent activity.

### 2. Data Cleaning

The dataset was prepared through standard preprocessing techniques, including:

- Missing value handling
- Duplicate removal
- Data validation
- Data type corrections
- Basic feature preparation

### 3. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand transaction behavior and identify potential fraud patterns by examining:

- Transaction distribution
- Customer spending behavior
- Fraud occurrence
- Regional activity
- Time-based trends

### 4. SQL Analysis

SQL queries were used to calculate business metrics and perform analytical tasks including:

- Fraud rate calculation
- Customer-level aggregation
- Transaction summaries
- Geographic analysis
- Risk-based segmentation
- KPI reporting

### 5. Dashboard Development

Interactive dashboards were created using Power BI and Tableau to present fraud insights through dynamic visualizations and filters.

### 6. Business Insights

Analytical findings were translated into practical recommendations that can support fraud monitoring, operational reporting, and business decision-making.